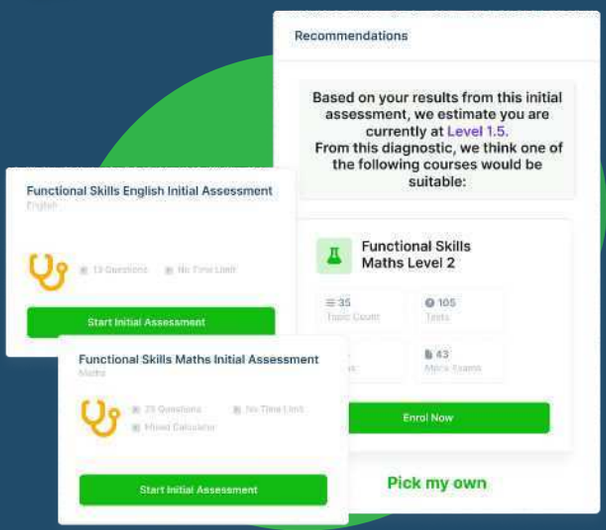
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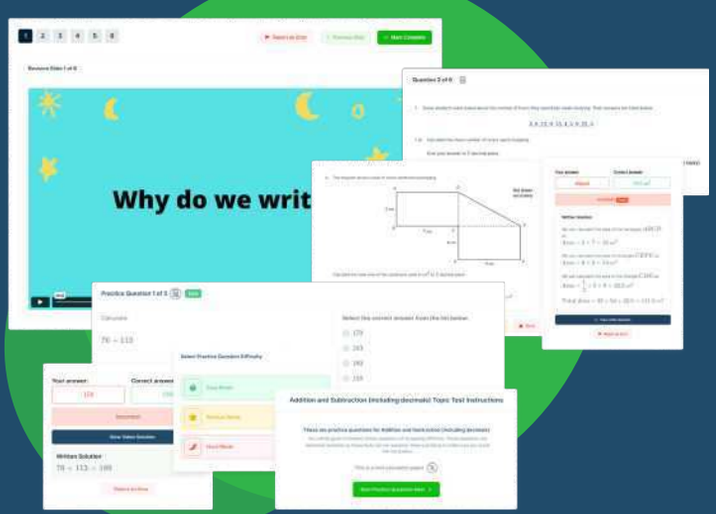
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Activity 2: Hotel

2 (a) Fiona is on a work placement in a hotel.

The hotel has a loyalty programme for the guests.

There are three million, four hundred thousand and nineteen guests currently signed up to the loyalty programme worldwide.

Write three million, four hundred thousand and nineteen as a number.

[1 mark]

Your answer: 3 400 019

2 (b) There are 216 guests currently staying at the hotel.

81 of these guests are signed up to the loyalty programme.

Fiona thinks that more than $\frac{4}{10}$ of these guests are signed up to the loyalty programme.

Is she correct?

Show how you decide.

[2 marks]

$\frac{81}{216} \xrightarrow{\div 9} \frac{9}{24} \xrightarrow{\div 3} \frac{3}{8} = \frac{15}{40}$

$\frac{15}{40} < \frac{16}{40}$ so

$\frac{4}{10} = \frac{16}{40}$ she's wrong.

Your answer: No $\frac{15}{40} < \frac{16}{40}$

Please turn over

Examiner

2 (c) The table below shows information about the 216 guests staying at the hotel on Sunday.

Sunday	
Number of nights stay	**Number of guests**
1	59
2	42
3	65
4	25
5	16
6	5
7	4
Total	216

Every Sunday and Monday, guests who will be staying for more nights than the mode are given a voucher for £10 off food and drink in the restaurant.

The total value of the voucners given to guests on Monday was ... total value of the vouchers given on Sunday.

What was the total value of the vouchers given on Monday?

[4 marks]

Sunday Mode = 3 nights (65)

25 + 16 + 5 + 4 = 50

50 x 10 = 500 £500 given on Sunday

Monday 4% increase

multiplier : 1.04

500 x 1.04 = £520

```
  500.00
x   1.04
--------
  20.00
+500.00
--------
  520
```

Your answer: £ 520

Please turn over

Examiner u

2 (d) Fiona uses this formula to work out the number of loyalty point[illegible] guest last year.

$$L = 300\left(s + \frac{r}{2}\right)$$

Where L = number of loyalty points collected in a year
s = number of nights stayed in a suite
r = number of nights stayed in a regular room

She then uses this table to put each guest into one of five tiers based on the number of loyalty points they earned.

Tier	Loyalty points collected in a year
1	0 – 599
2	600 – 1499
→ 3	1500 – 2999
4	3000 – 4999
5	5000+

Last year, Mrs Jones stayed 3 nights in a suite and 7 nights in a regular room.

Which tier should Fiona put Mrs Jones into?

You must show all your working.

[2 marks]

$L = 300\left(s + \frac{r}{2}\right)$

$= 300\left(3 + \frac{7}{2}\right)$

$= 1950$

Tier 3

$s = 3$

$r = 7$

$$\begin{array}{r} 300.0 \\ \times \quad 6.5 \\ \hline 150.0 \\ +\ 1800.0 \\ \hline 1950.0 \end{array}$$

Your answer: **Tier** 3

2 (e) A luxury room in the hotel has a bathtub 60.3 inches long.

What is the length of the bathtub in centimetres?

Use the conversion 1 inch = 2.54 cm

Give your answer correct to 1 decimal place.

[2 marks]

$60.3 \times 2.54 = 153.162$

$= 153.2$ to 1dp

$$\begin{array}{r} 60.30 \\ \times\ 2.54 \\ \hline 2.4012 \\ 30.150 \\ +\ 120.600 \\ \hline 153.8162 \end{array}$$

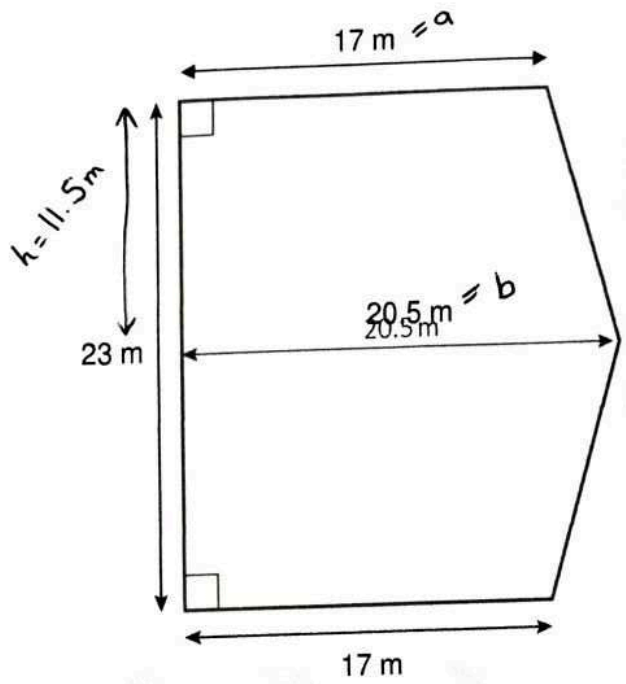
Your answer: 153.2 **cm**

Examiner use

2 (f) Fiona helps with organising an event at the hotel.

The diagram below shows the dimensions of the dance floor.

The shape of the dance floor has one line of symmetry.



Not drawn accurately

On the dance floor, each person must have at least 0.9 m^2 of floor space.

Work out the maximum number of people who can use this dance floor at one time.

[4 marks]

Area of trapezium = $\frac{1}{2}(a+b)h$

$\frac{1}{2} \times (17+20.5) \times 11.5 = 215.625$

$215.625 \times 2 = 431.25\ m^2$ whole area

$\frac{431.25}{0.9} = 479.166$

≈ 479 people

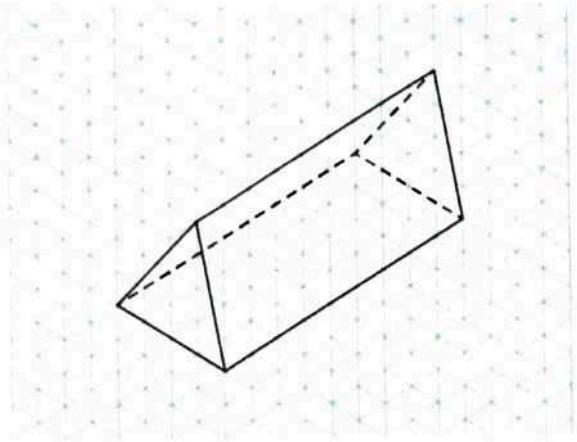
Your answer: 479 **people**

[Total marks: 15]

Activity 3: Product design

3 (a) Calista is a product designer at a toy factory.

The diagram shows a metal part in a toy robot.



What is the name of this shape?

Tick the correct answer.

- ☐ A pyramid
- ☐ B cylinder
- ☑ C prism
- ☐ D cuboid

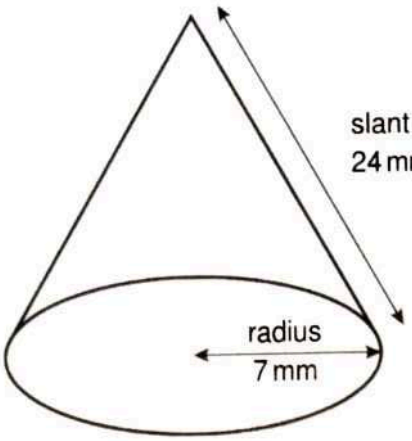
[1 mark]

Please turn over

Examiner use o

3 (b) Calista designs a plastic piece for a game.

The piece is in the shape of a cone.
The diagram shows the dimensions of the piece.



Not drawn accurately

Work out the surface area of the piece.

Use $A = \pi r^2 + \pi rs$
where A = surface area of a cone
r = radius of the circular base $= 7$ mm
s = slant height of the cone $= 24$ mm

Use $\pi = 3.14$

[2 marks]

$A = \pi(7)^2 + \pi(7)(24)$

$= 49\pi + 168\pi$

$= 217\pi$

$= 217 \times 3.14$

$= 681.38$

Your answer: 681.38 **mm²**

3 (c) Calista mixes three colours to make teal-coloured paint for the plastic game pieces.
She mixes blue, green and yellow paint in the ratio 4 : 2 : 1

Calista wants to make 14 litres of teal-coloured paint.
She thinks she needs 1.6 litres of yellow paint.

Is she correct?

Show how you decide.

[2 marks]

$14 \div (4 + 2 + 1) = 2$

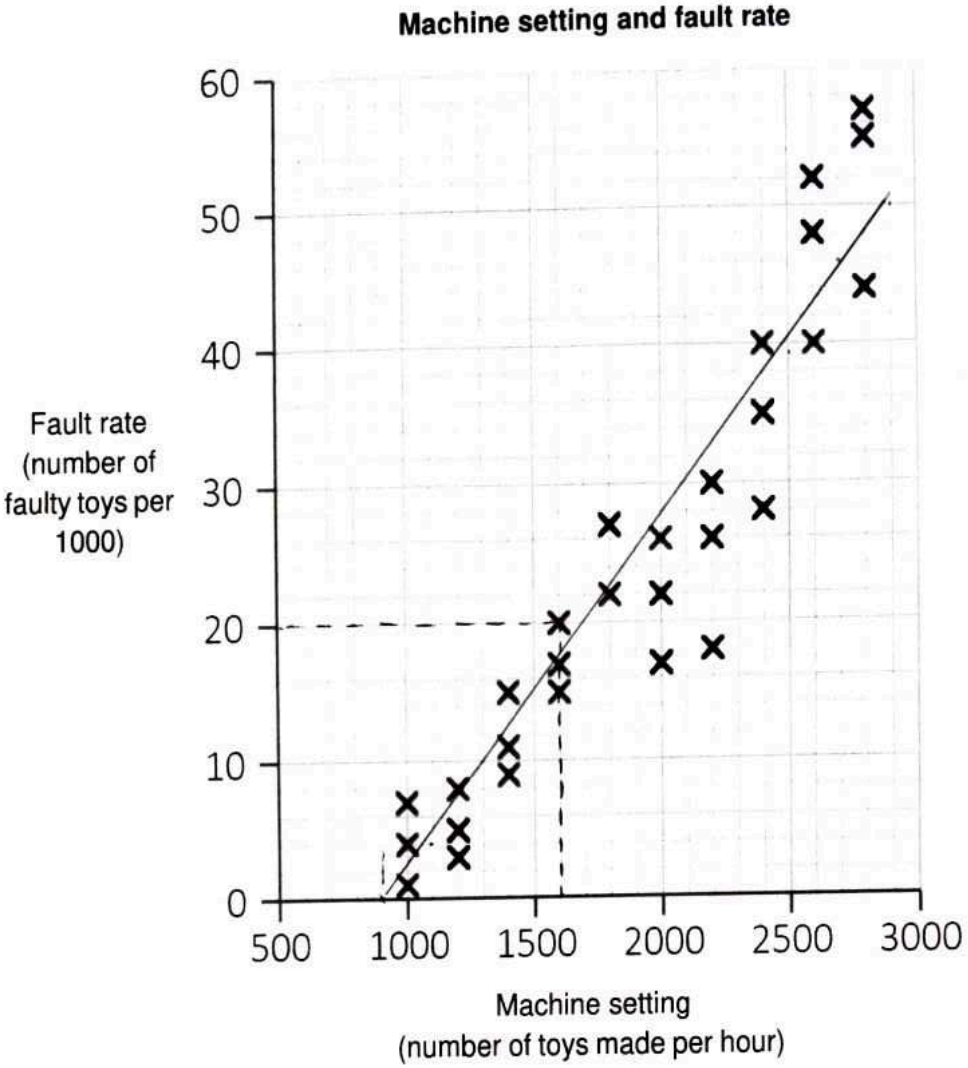
need 2 litres of yellow paint

Your answer:	No, need 2 litres of yellow

For use only

Examiner us

3 (d) The graph below shows the information about the settings and fault rate of a machine in the toy factory.



What was the highest fault rate when the machine was set to make 1600 toys per hour?

[1 mark]

See graph

Your answer: 20 **faulty toys per 1000**

Calista is writing a report.
She uses a line of best fit to find the fault rate when the machine setting is 2500

Calista writes, "The fault rate when the machine setting is 2500 as a percentage of the median fault rate when the machine setting is 2600 is more than 90%."

Is she correct?

Show how you decide.

[5 marks]

use only

Fault rate at 2500 is 41 (see line of best fit) (under Q3d)

Median at 2600 is 48

$\frac{41}{48} \times 100 = 85.4166\ldots\ \%$

No she's not correct the percentage is 85.4%.

Your answer:	No 85.4%.

Please turn over

Examiner

3 (f) Calista has this data about the volume of plastic used to make the toys she has designed.

Volume of plastic (p in cm^3) used to make a toy	Number of toys made	midpoints
$0 < p \leq 30$	26 000	15
$30 < p \leq 60$	42 000	45
$60 < p \leq 90$	17 000	75
$90 < p \leq 120$	9000	105
$120 < p \leq 150$	6000	135
Total:	100 000	

She works out an estimate of the mean volume of plastic used for each toy.

She wants to reduce this mean by 8%

If she is successful, what will the new mean be?

Give your answer correct to 1 decimal place.

[4 marks]

$26000 \times 15 = 390\,000$

$42000 \times 45 = 1890\,000$

$17000 \times 75 = 1\,275\,000$

$9000 \times 105 = 945\,000$

$6000 \times 135 = 810\,000$

$390\,000 + 1\,890\,000 + 1\,275\,000 + 945\,000 + 810\,000$

$= 5\,310\,000$

$\frac{5\,310\,000}{100\,000} = 53.1$ mean

reduced mean $53.1 \times 0.92 = 48.852$

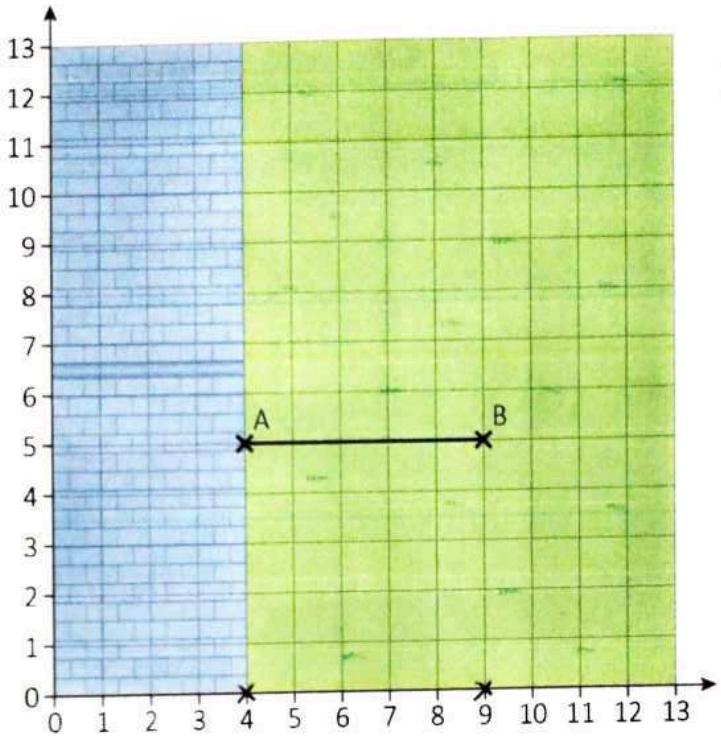
Your answer: 48.9

[Total marks: 15]

Activity 4: House extension

4 (a) Abdul wants to build an extension to his house.

The coordinate grid below shows the plan of his back garden where the extension will be.



Points A and B are two corners of the extension.
The line AB shows one wall of the extension.

The floor of the extension will be a square.

Write down a set of possible coordinates for the other two corners of the extension.

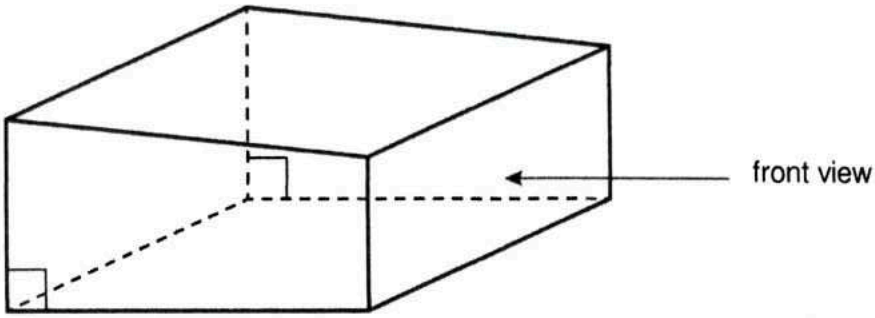
[2 marks]

Your answer: (4, 0) and (9, 0)

Please turn over

Examiner

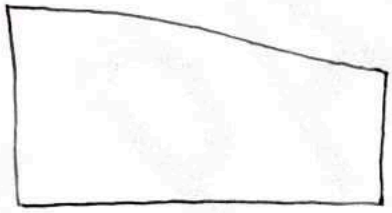
4 (b) The diagram shows the shape of the extension.



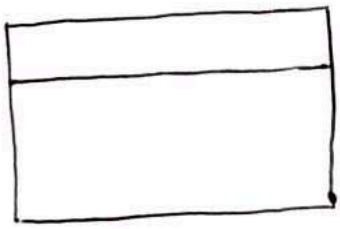
Draw the side view **and** the front view of the extension.

[2 marks]

Side view

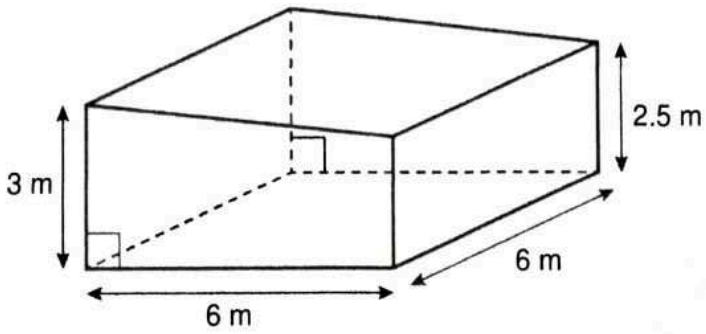


Front view



Examiner

4 (c) The diagram below shows the dimensions of the extension.



Not drawn accurately

Abdul wants to work out the power of the heater needed for the extension.

First, he needs to find the volume (V) of air inside the empty extension by using this formula:

$V = 0.5 \times w \times l \times (a + b)$

where a = greatest height (m)
b = smallest height (m)
w = width (m)
l = length (m)

Abdul then uses this formula to work out the power of the heater needed for the extension.

$P = 40V + 100n$

where P = power of the heater (watts)
V = volume of air inside the empty extension (m^3)
n = the number of windows

There will be 2 windows.

Work out the power of the heater needed for Abdul's extension.

[4 marks]

Volume = area of trapezium × 6

$= \frac{1}{2}(a+b) \times h \times 6$

$= \frac{1}{2} \times (3+2.5) \times 6 \times 6$

$= 99$

$P = 40 \times 99 + 100 \times 2$

$= 4160\,W$

Your answer: 4160 **watts**

Please turn over

Examine

...anning application.
He must include a scale drawing of the garden and the extension.

The width of the garden is 11.8 metres.
The drawing will use the scale 1 : 200

Abdul thinks that the width of the garden in the scale drawing is 5.4 cm long.

Is Abdul correct?

Show how you decide.

[2 marks]

11.8 x 100 = 1180 cm

1180 ÷ 200 = 5.9 cm

So No

Your answer:	No the width would be 5.9 cm

n about planning applications to hi

2020

Type of planning application	Received	Granted	Declined
Major	270	195	75
Residential	306	267	39
Commercial	124	94	30

Abdul thinks that, overall, residential applications are more likely to be granted than commercial applications.

Is he correct?

Show how you decide.

[2 marks]

residential $\frac{267}{306} = 0.8725...$

commercial $\frac{94}{124} = 0.75806...$

Your answer: Yes 87.3% are granted over 75.8% of commercial

Please turn over

Examiner u

4 (f) Abdul finds this information about planning applications to his local council in 2020

Type of planning application	Received	Granted	Declined
Major	270	195	75
Residential	306	267	39
Commercial	124	94	30

What is the probability that one of these applications chosen at random is a declined commercial application?

[1 mark]

$270 + 306 + 124 = 700$

$\frac{30}{700} = 0.0428\ldots$

Your answer: 0.04 or $\frac{7}{30}$

4 (g) Abdul is granted planning permission.

He does this calculation to work out how much concrete to buy:

$$\frac{(0.2 + 0.05) \times 6^2}{0.9}$$

What answer should Abdul get?

[2 marks]

BIDMAS

$\frac{(0.2+0.05) \times 6^2}{0.9} = \frac{(0.25) \times 36}{0.9}$

$= 10$

Your answer: 10

[Total marks: 15]

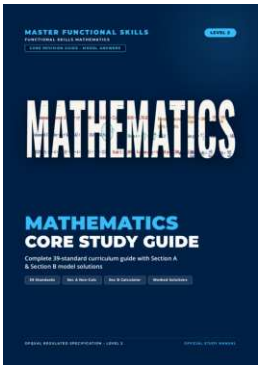
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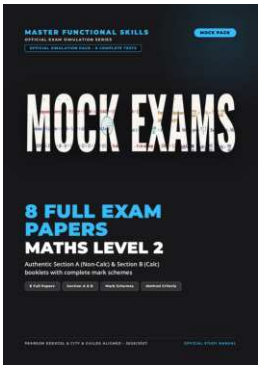


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